

AI-Driven Talent Acquisition Platform

HireLens AI

Transforming raw candidate profiles, resumes, and spreadsheets into structured, actionable hiring intelligence instantly.

Problem Statement vs. Our Solution

● The Hiring Challenge

- Recruiters spend hours manually parsing unstructured CVs and candidate profiles.
- Inconsistent evaluation metrics leading to suboptimal hiring choices.
- Spreadsheet clutter and data silos slowing down candidate pipelines.
- High vulnerability to human unconscious bias during early screening.

● The HireLens AI Solution

- Automated multi-format document extraction (PDF, DOCX, TXT, Excel, Images).
- Instant evaluation scores out of 10 with calibrated objective benchmarks.
- Structured identification of top core strengths, gaps, and hidden risks.
- Secure portal access with custom OpenAI API integration for secure analysis.

Technical Architecture & Workflow

Technology Stack

- **Frontend UI:** Streamlit framework with reactive session states.
- **AI Core:** LangChain orchestration coupled with OpenAI GPT-4o-mini.
- **Parsing Libraries:** PyPDF, Python-DocX, and Pandas for data frames.
- **OCR Pipeline:** Tesseract-OCR engine combined with Pillow imaging.

Processing Pipeline

- **1. Ingestion:** Secure file upload interface supporting 9+ document formats.
- **2. Extraction:** Normalization of raw text data and tabular string structures.
- **3. Evaluation:** Role-specific system prompts tailored for expert HR analysts.
- **4. Output Delivery:** Clean markdown reports featuring metrics and actionable cues.

IMPACT & METRICS

Efficiency & Performance Gains

85%

Reduction in Initial Screening Time

10x

Faster Spreadsheet & Data Parsing

100%

Structured & Objective Scoring



Strategic Value for Recruitment Teams

By replacing tedious manual reviews with intelligent automated processing, recruiting teams can reallocate hours toward high-value candidate engagement, deep tech interviews, and faster overall time-to-hire.

Future Scope & Scaling Outlook



Upcoming Enhancements

- Automated contextual interview question generation targeting candidate gaps.
- Multi-candidate comparative ranking and filtering dashboard view.
- Seamless native integrations with major ATS platforms (Workday, Greenhouse).
- Vector embedding search for historical talent pools and past applicant tracking.



Project Conclusion

- HireLens AI successfully bridges the gap between unstructured data and data-driven hiring.
- Offers a modular, highly secure architecture ready for enterprise deployment.
- Empowers HR professionals with lightning-fast insights while ensuring complete data safety.